

General-Purpose Technologies and Stock Market Bubbles¹

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NFA Conference
September 26, 2026

¹Link to paper: <https://arxiv.org/abs/2501.08215> 

Technological innovation and bubbles

- Many asset-price booms coincide with major technological innovations (Scheinkman, [2014](#); Quinn and Turner, [2020](#))
- Examples:
 - 1720s Mississippi and South Sea bubbles: Atlantic trade, insurance
 - 1840s Railway Mania: steam engine, railway network
 - 1890s Bicycle Mania: pneumatic tire
 - 1920s US stock boom: electricity, automobiles, radio
 - 1990s dot-com bubble: Internet; today: AI?
- General-purpose technologies (GPTs) (Bresnahan and Trajtenberg, [1995](#)):
 - (pervasiveness) useful across many sectors
 - (improvement) improve over time
 - (innovation spanning) make it easier to develop further innovations

Rational bubble despite anticipated collapse

- Bubble: asset price (Q) > fundamental value (V)
 - V = expected present discounted value of dividends (D)
- “Rational bubble”: $Q > V$ despite rational agents with common beliefs and information (Hirano and Toda, 2024)
- Standard difficulties:
 - Dividends tighten no-bubble condition relative to pure bubble ($D \equiv 0$)
 - Known collapse date rules out bubbles by backward induction
- Can bubble be necessary despite anticipated complete collapse?

This paper

- Macro-finance model of innovation and stock price bubble
- Skilled agents choose between production and R&D
- R&D creates new firms whose shares pay dividends
- Knowledge spillovers differ across production factors
 - Emerging GPT: unbalanced growth
 - Mature GPT: absorbing balanced-growth state
- Agents rationally anticipate regime transition and bubble collapse

Main results

1. **Bubble necessity despite anticipated collapse**
 - Unique equilibrium has $Q_t > V_t$ during unbalanced growth
 - Survival probability affects duration, not existence
2. **Bubble-backed innovation and regime transition**
 - Capitalizing bubble into IPO proceeds increases R&D employment
 - Later collapse implies larger contraction but more inherited knowledge
3. **Balanced growth: knife-edge**
 - Restrictions generating balanced growth also rule out bubble by construction

Related literature

- **Rational bubbles:** Samuelson (1958), Bewley (1980), Tirole (1985), Scheinkman and Weiss (1986), Kocherlakota (1992), and Santos and Woodford (1997)
- **Bubbles attached to dividend-paying assets:** Hirano and Toda (2024, 2025a,b) and Sorger (2026)
- **Collapsing bubbles:** Blanchard (1979), Weil (1987), Michel and Wigniolle (2003), and Guerron-Quintana et al. (2023)
- **Technological innovation and asset booms:** Olivier (2000), Zeira (1999), Pástor and Veronesi (2009), and Haddad et al. (2022)

Endowment economy

- Two-period OLG, $t = 0, 1, \dots$
- Unit mass of young agents born each date
- Young receive endowment $e_t > 0$, old receive none
- Initial old own one unit of long-lived asset
 - ex-dividend price $Q_t > 0$
 - dividend $D_t > 0$
- Complete markets with Arrow securities in zero net supply
- Adapted stochastic process (e_t, D_t)
- Utility is log Epstein–Zin (EIS = 1, $\beta \in (0, 1)$, $\gamma \geq 0$):

$$U(c_t^y, c_{t+1}^o) = (1 - \beta) \log c_t^y + \beta \log E_t[(c_{t+1}^o)^{1-\gamma}]^{\frac{1}{1-\gamma}}$$

Equilibrium

- Generation- t budget constraints:

$$c_t^y + Q_t \theta_t = e_t,$$

$$c_{t+1}^o = (Q_{t+1} + D_{t+1}) \theta_t$$

- Market clearing:

$$c_t^y + c_t^o = e_t + D_t,$$

$$\theta_t = 1$$

- Unit EIS implies $c_t^y = (1 - \beta)e_t$

► Solution

Unique equilibrium

- Young's budget constraint and asset-market clearing imply

$$Q_t = e_t - c_t^y = \beta e_t$$

Proposition

Unique rational expectations equilibrium exists, with asset price

$$Q_t = \beta e_t,$$

and consumption

$$(c_t^y, c_t^o) = ((1 - \beta)e_t, \beta e_t + D_t).$$

Fundamental value and bubble

- Let one-period SDF be $m_{t \rightarrow t+1}$,
 $m_{t \rightarrow T} := m_{t \rightarrow t+1} \times \cdots \times m_{T-1 \rightarrow T}$ ► SDF
- Euler equation:

$$Q_t = E_t[m_{t \rightarrow t+1}(Q_{t+1} + D_{t+1})]$$

- Iteration and monotone convergence yield

$$Q_t = E_t \left[\underbrace{\sum_{s=t+1}^{\infty} m_{t \rightarrow s} D_s}_{=: V_t} \right] + \underbrace{\lim_{T \rightarrow \infty} E_t[m_{t \rightarrow T} Q_T]}_{=: B_t}$$

- Hence $Q_t = V_t$ if and only if **no-bubble condition** holds:

$$\lim_{T \rightarrow \infty} E_t[m_{t \rightarrow T} Q_T] = 0$$

Assumptions: Temporary unbalanced growth

1. Two states $z_t \in \{u, b\}$ with transition probabilities

$$\Pr(z_{t+1} = u \mid z_t = u) = \pi \in (0, 1),$$

$$\Pr(z_{t+1} = b \mid z_t = b) = 1.$$

2. If state b is reached at τ , all subsequent endowments and dividends are deterministic, with

$$\frac{e_{t+1}}{e_t} = \frac{D_{t+1}}{D_t}, \quad t \geq \tau.$$

3. For every $t \geq 1$, past regime history determines two possible pairs (e_t^u, D_t^u) and (e_t^b, D_t^b) . Current state selects realized pair; no other aggregate uncertainty.
- u : endowments and dividends may grow at different rates
 - b : absorbing balanced-growth state

Bubble characterization under anticipated collapse

Let $\Pi_t^z := Q_t^z + D_t^z$ denote stock payoff conditional on state $z \in \{u, b\}$

Theorem

Under maintained assumptions,

1. *There is no bubble ($Q_t = V_t$) if $z_t = b$.*
2. *If $z_0 = u$, bubble exists ($Q_0 > V_0$) if and only if*

Vanishing dividend yield:

$$\sum_{t=1}^{\infty} \frac{D_t^u}{Q_t^u} < \infty,$$

Large transition-state decline:

$$\sum_{t=1}^{\infty} \left(\frac{\Pi_t^b}{\Pi_t^u} \right)^{1-\gamma} < \infty.$$

Economic interpretation

- Suppose $\gamma < 1$
 1. During state u : dividend yield tends to zero, price-dividend ratio to infinity
 2. At transition, consumption and stock payoff fall relative to continuation path
 3. Payoff decline dominates increased marginal valuation
- Bubble survives state u only if eventual collapse becomes sufficiently severe
 - $\gamma = 1$: each second-series term equals one
 - $\gamma > 1$: transition-state terms diverge when $\Pi_t^b / \Pi_t^u \rightarrow 0$
- Similar result with general Epstein–Zin [▶ Details](#)

Model with innovation and intangible capital

- Variety-expansion model with monopolistic competition (Grossman and Helpman, 1991)
- Mass $L > 0$ of unskilled agents
 - work in final-good sector
- Mass $H > 0$ of skilled agents
 - produce knowledge-intensive intermediate goods, or
 - conduct R&D and establish new firms
- Each intermediate-good firm issues one share and pays monopoly profits as dividends
- n_t : variety count, knowledge stock, and stock supply

Equilibrium

Let $\zeta_t := A_{Xt}H/(A_{Lt}L)$ and $g(x) := (F_X/F_L)(x, 1)$

Proposition

If innovation occurs, $\phi_t \in (0, 1)$ uniquely solves

$$\frac{1}{aH} = \phi_t - 1 + \beta + \frac{\beta\sigma}{(\sigma - 1)\zeta_t g(\zeta_t \phi_t)}.$$

Prices are

$$\begin{aligned} P_t &= p_t(j) = F_X A_{Xt}, \\ w_{Ht} &= \frac{\sigma - 1}{\sigma} F_X A_{Xt}, & w_{Lt} &= F_L A_{Lt}, \\ Q_t &= \frac{w_{Ht}}{an_t}. \end{aligned}$$

Knowledge spillovers

Assumption

Elasticity of substitution (between knowledge-intensive good and unskilled labor) satisfies $1/\rho < 1$, and factor-augmenting productivities satisfy

$$(A_{Xt}, A_{Lt}) = (A_X n_t^{\xi_{zt}}, A_L n_t^{\lambda_{zt}})$$

with

$$\xi_u > \lambda_u > \lambda_b = \xi_b.$$

- State u : uneven factor-augmenting spillovers and unbalanced growth
- State b : equal spillovers and balanced growth
- If $1/\rho < 1$, then $\kappa > 0$ is equivalent to $\xi_u > \lambda_u$

Innovation and knowledge growth

With CES production, equilibrium condition becomes

$$\frac{1}{aH} = \phi_t - 1 + \beta + \frac{\beta\sigma(1-\alpha)}{(\sigma-1)\alpha} \left(\frac{A_X H}{A_L L} \right)^{\rho-1} n_t^{(\xi_{z_t} - \lambda_{z_t})(\rho-1)} \phi_t^\rho$$

Proposition

Conditional on continued state u , $\phi_t \downarrow 0$ and

$$\frac{n_{t+1}}{n_t} \rightarrow G_u := 1 + aH.$$

In state b , $\phi_t = \phi_b$ and

$$\frac{n_{t+1}}{n_t} = G_b := 1 + a(1 - \phi_b)H < G_u.$$

Inevitable stock price bubbles

Given active innovation,

$$\frac{D_t}{Q_t} = \frac{aH}{\sigma - 1} \phi_t$$

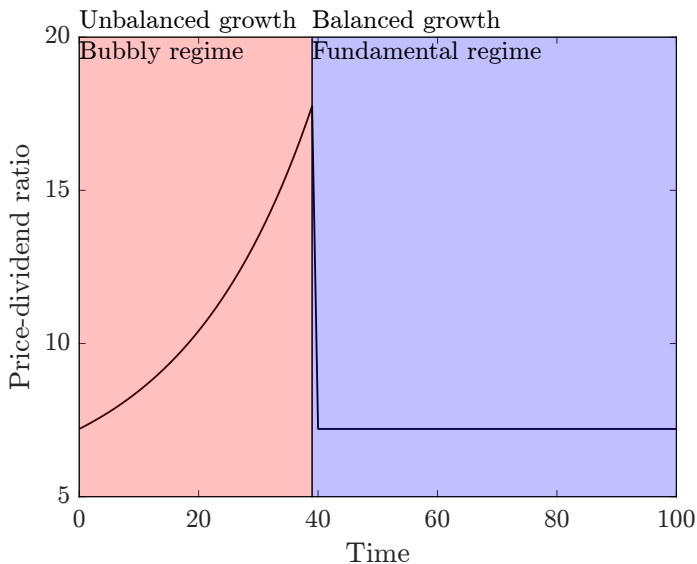
Theorem

Suppose production is CES, innovation persists throughout state u , and $\gamma < 1$. Unique equilibrium:

1. *If $z_t = u$, $Q_t > V_t$ and Q_t/D_t grows asymptotically at exponential rate*
2. *If $z_t = b$, $Q_t = V_t$ and Q_t/D_t is constant*

Agents rationally anticipate transition to b and complete bubble collapse

Price-dividend ratio dynamics



Bubble-backed IPO proceeds

At fixed state- u history, write $Q_t = V_t + B_t$ and let innovators retain

$$Q_t^{\text{IPO}}(\chi) = V_t + \chi B_t, \quad \chi \in [0, 1]$$

Proposition

As χ increases:

- $\phi_t(\chi)$ decreases
- *R&D employment and next-period knowledge increase*
- *skilled wage and skill premium increase*
- *current output and unskilled wage decrease*

One-period partial-equilibrium incidence experiment; all shares continue trading at Q_t

► Details

Regime-transition consequences

Suppose economy switches from u to b at date t

Proposition

As transition occurs later,

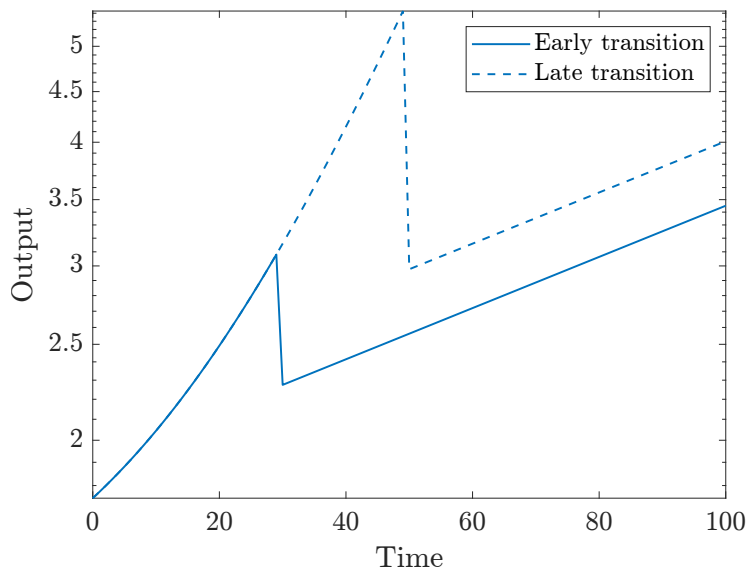
$$\frac{Y_t^b}{Y_{t-1}^u} \sim n_t^{\lambda_b - \lambda_u} \rightarrow 0, \quad \frac{Q_t^b}{Q_{t-1}^u} \sim n_t^{\lambda_b - \lambda_u} \rightarrow 0.$$

If first transition occurs at τ , then for every fixed $k \geq 0$,

$$Y_{\tau+k} = C_Y G_b^{\lambda_b k} n_\tau^{\lambda_b},$$
$$w_{i,\tau+k} = C_i G_b^{\lambda_b k} n_\tau^{\lambda_b}, \quad i \in \{H, L\}.$$

Later transition implies larger contraction but higher post-transition output and wage levels; post-transition skill premium unchanged

Output dynamics



Conclusion

- Temporary unbalanced growth can make collapsing bubble on dividend-paying stocks necessary
- Uneven GPT spillovers generate exponentially growing price-dividend ratio
- Bubble-backed IPO proceeds increase R&D and next-period knowledge in incidence experiment
- Later collapse more severe but leaves larger productive knowledge stock
- Balanced growth imposes knife-edge restrictions with strong asset-pricing consequences

▶ Infinite horizon

▶ Bezos

Optimal consumption with log utility

- Let $R_{t+1} = (Q_{t+1} + D_{t+1})/Q_t$ be gross return
- Saving problem equivalent to

$$\max_{0 < c < e_t} (1 - \beta) \log c + \beta \log(e_t - c) + \text{constant}$$

- Hence

$$c_t^y = (1 - \beta)e_t, \quad Q_t = \beta e_t$$

◀ Endowment economy

Stochastic discount factor

- One-period SDF:

$$m_{t \rightarrow t+1} = \frac{\beta}{1 - \beta} \frac{c_t^y (c_{t+1}^o)^{-\gamma}}{E_t[(c_{t+1}^o)^{1-\gamma}]}$$

- Using equilibrium consumption,

$$\begin{aligned} m_{t \rightarrow t+1} &= \frac{\beta}{1 - \beta} \frac{(1 - \beta)e_t(\beta e_{t+1} + D_{t+1})^{-\gamma}}{E_t[(\beta e_{t+1} + D_{t+1})^{1-\gamma}]} \\ &= \frac{Q_t(Q_{t+1} + D_{t+1})^{-\gamma}}{E_t[(Q_{t+1} + D_{t+1})^{1-\gamma}]} \end{aligned}$$

◀ Bubble theorem

Bubble-characterization proof

- Define

$$S_t := \pi^t E_0[m_{0 \rightarrow t} Q_t \mid \tau > t]$$

- Equilibrium SDF substitution gives

$$S_t = Q_0 \prod_{s=1}^t \frac{1}{1 + D_s^u / Q_s^u + \frac{1-\pi}{\pi} (\pi_s^u / Q_s^u) (\pi_s^b / \pi_s^u)^{1-\gamma}}$$

- Writing denominator as $1 + a_s$, inequalities

$$1 + a \leq e^a, \quad \prod_{s=1}^t (1 + a_s) \geq 1 + \sum_{s=1}^t a_s$$

- Hence $\lim_t S_t > 0$ if and only if $\sum_s a_s < \infty$, equivalent to theorem's two conditions

General EIS

- $\mathcal{R}_t(X)$: conditional CRRA certainty equivalent for positive X

$$\mathcal{R}_t(X) = \begin{cases} E_t[X^{1-\gamma}]^{\frac{1}{1-\gamma}} & \text{if } \gamma \neq 1 \\ \exp(E_t[\log X]) & \text{if } \gamma = 1 \end{cases}$$

- Let $\psi \neq 1$ denote EIS and define $x_t = Q_t/e_t$, $d_t = D_t/e_t$, and $G_{t+1} = e_{t+1}/e_t$; then

$$\frac{x_t^\psi}{1 - x_t} = \left(\frac{\beta}{1 - \beta} \right)^\psi [\mathcal{R}_t(G_{t+1}(x_{t+1} + d_{t+1}))]^\psi$$

- Under bounded growth and uniform interiority:
 - uniformly interior equilibrium exists
 - same primitive relative-growth conditions characterize bubbles
 - complete collapse still requires $\gamma < 1$
- For $\psi > 1$, greatest equilibrium is unique uniformly interior equilibrium; other equilibria may approach zero-price boundary

No-innovation corner

- Innovation occurs if

$$\frac{1}{aH} < \beta + \frac{\beta\sigma}{(\sigma-1)\zeta_t g(\zeta_t)}$$

- If condition fails, $\phi_t = 1$ and

$$Q_t = \frac{\beta(Hw_{Ht} + Lw_{Lt})}{n_t}$$

- Either way, ϕ_t depends on time only through relative effective productivity

$$\zeta_t = \frac{A_{Xt}H}{A_{Lt}L}$$

Full-model bubble-necessity proof

- When innovation occurs,

$$\frac{D_t}{Q_t} = \frac{aH}{\sigma - 1} \phi_t$$

- In state u , ϕ_t decreases at exponential rate, so

$$\sum_t \frac{D_t^u}{Q_t^u} < \infty$$

- At transition,

$$\frac{Q_t^b}{Q_t^u} \sim n_t^{\lambda_b - \lambda_u} \rightarrow 0$$

- Together with $\gamma < 1$, this verifies

$$\sum_t \left(\frac{\Pi_t^b}{\Pi_t^u} \right)^{1-\gamma} < \infty$$

- Endowment-economy theorem then implies $Q_t > V_t$ in state u

Bubble capture and occupational choice

- Holding n_t , (A_{Xt}, A_{Lt}) , V_t , and B_t at benchmark values, occupational choice satisfies

$$\begin{aligned} a n_t Q_t^{\text{IPO}}(\chi) &\leq w_{Ht}(\phi_t(\chi)), \\ (1 - \phi_t(\chi)) [w_{Ht}(\phi_t(\chi)) - a n_t Q_t^{\text{IPO}}(\chi)] &= 0 \end{aligned}$$

- Unique $\phi_t(\chi)$ decreases with χ , so

$$\tilde{n}_{t+1}(\chi) = [1 + aH(1 - \phi_t(\chi))] n_t$$

increases with fraction of bubble component captured by innovator

- Comparison excludes financial-market clearing, wealth effects, and fundamental-value recomputation

Infinitely lived agents

Use demographic structure from Weil (1989):

- every cohort lives forever
- new skilled and unskilled cohorts enter at each date
- agents supply labor only at birth, then consume from financial wealth
- production, innovation, and regime process unchanged
- Recursive utility is

$$U_{i,\tau,t} = c_{i,\tau,t}^{1-\beta} E_t[U_{i,\tau,t+1}^{1-\gamma}]^{\frac{\beta}{1-\gamma}}$$

- Unit EIS implies each agent consumes fraction $1 - \beta$ of wealth

[◀ Return](#)[▶ Result](#)

Bubble necessity with infinitely lived agents

Theorem

Under active innovation and admissible continuation values, aggregate allocation, stock price, continuation-value solution, and SDF are unique. Furthermore:

1. *In state u , $Q_t > V_t$ and Q_t/D_t grows asymptotically at exponential rate*
 2. *In state b , $Q_t = V_t$ and Q_t/D_t is constant*
- Each agent satisfies individual transversality condition
 - Existing cohorts run down stock positions; new cohorts replenish aggregate stock demand with labor income
 - Finite lives unnecessary; continual cohort entry remains essential

Industrial bubbles

“The ones that are industrial are not nearly as bad. They could even be good because when the dust settles and you see who are the winners, society benefits from those inventions.”

Jeff Bezos, Italian Tech Week, October 3, 2025

[Video recording](#)

◀ Main results

◀ Conclusion

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